

An introduction to Quantum Key Distribution (QKD): The BB84 Protocol

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The BB84 Protocol

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1. Abstract

Quantum Key Distribution (QKD) is an example of post-quantum cryptography. A post-quantum cryptographic algorithm is an algorithm that is believed to be secure even when an attacker is able to use quantum computers, which can utilize significantly higher computational capability than classical computers, in order to break a cryptosystem. These algorithms are being developed in order to prepare for a future where quantum computers are commonplace. Many cryptographic schemes in use today depend on their security based on mathematical problems such as:

Prime Factorization, the Discrete Log Problem, etc.,

which can all be easily solved by a quantum computer. As a result, these cryptosystems which are commonly used in today's world will become obsolete.

This report will be discussing the implementation of BB84, a quantum key distribution protocol that utilizes properties of quantum mechanics in order to distribute keys, and is immune to the security issues faced by classical cryptographic schemes.

2. BB84 Motivation

BB84 is only a key distribution protocol, and is typically used with a cryptosystem such as the One Time Pad. A brief description of the One Time Pad is as follows:

One Time Pad (OTP):

1. **Encryption:** Alice encrypts messages M using key K_A , so Alice sends ciphertext $C = M \oplus K_A$ ($\oplus$ is binary addition. Thus, $0+0=1+1=0$, $0+1=1+0=1$)
2. **Transmission:** The ciphertext C is sent to Bob over a public channel
3. **Decryption:** Bob decrypts the message using his key K_B , so Bob gets the message $M' = C \oplus K_B$ (by again, using binary addition. The two keys of Bob and Alice are the same in OTP)

OTP is information-theoretically secure. This means that an adversary, Eve, cannot get any information about the message that was sent by intercepting the ciphertext. To be secure, OTP's key must be:

1. Random
2. At least as long as the message
3. Never reused.

4. Kept completely secret

What we want is an ideal way to generate keys for Alice and Bob, K_A and K_B , such that if it is detected that eavesdropper Eve has obtained too much information, the key distribution protocol aborts.

Properties of such a key:

1. **Correctness:** The probability that Alice and Bob's key differ but the protocol does not abort is small:

$$\Pr[K_A \neq K_B] \leq \epsilon$$

Where ϵ is some small arbitrary constant.

2. **Close to a perfect key:**
 - a. Eve does not have knowledge of the key
 - b. Individual key bits are uncorrelated

The motivation of developing a protocol, such as the BB84 Protocol, is to generate keys which have the same security properties that the OTP algorithm desires. We will see how the protocol is able to generate keys which can easily detect when an adversary Eve intercepts the key distribution protocol.

3. Prerequisites

BB84 (and other QKD algorithms) use fundamental properties of quantum mechanics as a part of the protocol. We will briefly mention the two important properties, Heisenberg's Uncertainty Principle and quantum superposition, in the extent to which we need to understand them in order to follow the protocol. The BB84 protocol typically uses photons for sending encoded bits, so we will also exclusively mention photons in this report. However, it is important to note that the following phenomena are observable for any particles small enough to exhibit these quantum properties.

3.1. Heisenberg's uncertainty principle

The following principle is applicable to two properties which are conjugate variables (the classical example consists of properties of position and momentum, or time and energy). Conjugate variables are essentially two variables or properties where measuring one results in losing the information about the other. We will first demonstrate what the uncertainty principle is in general before introducing two

conjugate variables which will be used for the BB84 protocol. We will now establish how the following principle functions (for the context of a BB84 instance).

Consider two binary properties of a photon, A and B . Since the two properties are binary, let's define the two possible states of A as a^+ and a^- , and let's define the two binary properties of B as b^+ and b^- . Now assume that we have an apparatus (or filter) which determines the state of property A , and another apparatus which determines the state of property B . The probability distribution of the properties states is dependent on their probability amplitudes, but for the sake of this report we will use the probability distribution used in the BB84 protocol. Thus, the probabilities for the states are:

$$\begin{aligned} Pr[A = a^+] &= Pr[A = a^-] = \frac{1}{2} \\ Pr[B = b^+] &= Pr[B = b^-] = \frac{1}{2} \end{aligned}$$

The first observation to make is that the apparatus is reliable. Meaning that if we measure a property to be in a specific state, then measuring it again will result in that state with a probability of 1. Thus, we get:

If property A is measured to be in state a^+ , then measuring property A again will return a^+ with $Pr[A = a^+] = 1$

The second observation to note is that measuring property A and then property B results in the same probability distribution for property B :

$$\begin{aligned} \text{If } A = a^+ \text{ then } Pr[B = b^+] &= Pr[B = b^-] = \frac{1}{2} \\ \text{If } A = a^- \text{ then } Pr[B = b^+] &= Pr[B = b^-] = \frac{1}{2} \end{aligned}$$

The interesting observation is when we measure property A , then measure property B , and then measure property A again. What we find here is that property A has “returned” to the probability distribution of $\frac{1}{2}$ for each state:

$$\begin{aligned} \text{If } A = a^+ \text{ and then, } B = b^+, \text{ then we find that } Pr[A = a^+] &= Pr[A = a^-] = \frac{1}{2} \\ \text{If } A = a^+ \text{ and then, } B = b^-, \text{ then we find that } Pr[A = a^+] &= Pr[A = a^-] = \frac{1}{2} \\ \text{If } A = a^- \text{ and then, } B = b^+, \text{ then we find that } Pr[A = a^+] &= Pr[A = a^-] = \frac{1}{2} \\ \text{If } A = a^- \text{ and then, } B = b^-, \text{ then we find that } Pr[A = a^+] &= Pr[A = a^-] = \frac{1}{2} \end{aligned}$$

Notice that in every scenario, it appears that measuring one of the properties results in the state of the other property to end up in uncertainty. This phenomenon is known as Heisenberg's uncertainty principle and it will be used in the BB84 protocol.

3.2. Quantum Superposition

It is not necessary to understand superposition to follow the BB84 protocol, so we will not explain it in detail. However, understanding it to a basic level is helpful for understanding how qubits (or quantum bits) are represented. We only need to know that superposition refers to the fundamental principle that a quantum property can exist in different quantum states at the same time. For an example, if we are to use our explanation from Heisenberg's uncertainty principle as context, then we can state that:

After determining property B to be b^+ , property A is in a superposition of being in states a^+ and a^- .

What this essentially means is that property A exists in both states after measuring property B .

We will discuss the representation of superposition as we discuss qubits. Although the protocol does use qubits, it is possible to present the protocol without the use of qubits, and instead with simplifications. However, we will discuss qubits first in order to have a full understanding of the protocol in mathematical notation.

4. Qubits

In order to analyse the protocol and properties of Quantum Key Distribution protocols, we must introduce the concept of qubits. Qubits are quantum states which are used to encode information in a physical system. The property used to represent the bits are chosen to be binary. We will first encode physical bits (0's and 1's) to a selectively chosen property of photons first before analysing the representation of superposition among the qubits.

4.1. Bit Encoding in BB84

In order to use the BB84 protocol, we must first find a way to encode classical bits to some property of photons.

For Protocol, we can use the polarization of photons to encrypt bits into quantum states using the following two distinct basis of the oscillation property of photons:

- Rectilinear Basis: A photon will either oscillate at $\leftrightarrow$ (0°) or $\updownarrow$ (90°)

- Diagonal Basis: A photon will either oscillate at $\nearrow$ (45°) or $\nwarrow$ (-45°)

The reason we have selected these two distinct basis is because they function like property A and property B from Heisenberg's uncertainty principle explanation. If we simply substitute property A for a photon's oscillation along the rectilinear basis, and substitute property B for a photon's oscillation along the diagonal basis, we end up in the same scenario. With this substitution, we can see that A is binary with the states a^+ : $\leftrightarrow$ (0°), and a^- : $\updownarrow$ (90°), and B is binary with the states b^+ : $\nearrow$ (45°), and b^- : $\nwarrow$ (-45°).

Assume that we have a polarization filter which is designed to measure photons in the rectilinear basis. In such a case, if the photon is polarized to be, $\leftrightarrow$ (0°), then it will be measured to be $\leftrightarrow$ (0°) certainly (with 100% probability), and if the photon is polarized to be $\updownarrow$ (90°), then it will be measured to be $\updownarrow$ (90°) certainly (with 100% probability).

What happens when we send a photon encoded in the diagonal basis through this filter? In such a case, the photon will be measured to be $\nearrow$ (45°) at 50% probability, and $\nwarrow$ (-45°) at 50% probability. As a result, the photon exiting the filter will have the polarization as $\leftrightarrow$ (0°) or $\updownarrow$ (90°), losing the bit data that was encoded in the original diagonal basis.

Thus, it is important to notice that measuring the diagonal basis of a photon that is encoded in the rectilinear basis will not yield information about the encoded data (and vice versa). This phenomenon is important to understanding how the BB84 protocol works.

Now, we must choose a convention for encoding bits.

Let us choose the following bit encoding for our protocol:

- **Rectilinear Basis (aka Computational Basis) represented as C :**
 - 0 is encoded to $\leftrightarrow$ (0°)
 - 1 is encoded to $\updownarrow$ (90°)
- **Diagonal Basis (aka Hadamard Basis) represented as H :**
 - 0 is encoded to $\nearrow$ (45°)
 - 1 is encoded to $\nwarrow$ (-45°)

4.2. Quantum State Representation

Now that we have discussed how to encode bits to quantum states, we will now represent the general quantum state of a property Ψ with Dirac notation: $|\Psi\rangle$. In terms of qubits, $|\Psi\rangle$ is in superposition of being in state $|0\rangle$ and state $|1\rangle$, which is a wave function represented by:

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α and β are the probability amplitudes of their respective qubit states. Since α and β are the probability amplitudes, we determine the probabilities of the qubit states from $|\alpha|^2$ and $|\beta|^2$. Thus, this normalization condition requires that the probabilities sum up to 1:

$$|\alpha|^2 + |\beta|^2 = 1$$

To represent the vectors associated with a qubit's state when in a superposition, we use vectors in Hilbert spaces. A Hilbert space H is a vector space over the complex numbers $\mathbb{C}$. Vectors are represented as $|x\rangle \in H$. Hilbert spaces are used to describe the probability of the qubit state of a qubit. For an example, for a qubit $|\Psi\rangle$ which can either have the qubit state $|0\rangle$ or $|1\rangle$, we can represent the probability of measuring the qubit in a specific state as:

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Thus, the two possible states can be represented by the following vectors in the 2-dimensional Hilbert space (We use a 2-dimensional Hilbert space because a qubit can only be measured to have one of two states):

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

For the BB84 protocol, since we have 50% probability of measuring each property to be in a specific binary state, we get that:

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\Psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

as we can see, $\alpha = \frac{1}{\sqrt{2}} \Rightarrow \text{Pr}(|0\rangle) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

$$\text{and } \beta = \frac{1}{\sqrt{2}} \Rightarrow Pr(|1\rangle) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

therefore, we have that $Pr(|0\rangle) + Pr(|1\rangle) = \frac{1}{2} + \frac{1}{2} = 1$ as required.

Now, we can represent a qubit Ψ as being in one of four states (two from each basis) like so:

- **Rectilinear Basis (aka Computational Basis) represented as C :**
 - $0, \leftrightarrow (0^\circ)$, is represented as $|\Psi_{00}\rangle$
 - $1, \updownarrow (90^\circ)$, is represented as $|\Psi_{10}\rangle$
- **Diagonal Basis (aka Hadamard Basis) represented as H :**
 - $0, \swarrow (45^\circ)$, is represented as $|\Psi_{01}\rangle$
 - $1, \searrow (-45^\circ)$, is represented as $|\Psi_{11}\rangle$

5. BB84 Protocol

We will now discuss the BB84 Protocol and show the steps that are taken to implement it.

In the setup for the protocol, Alice and Bob send the quantum states of the photons through a quantum channel. Alice and Bob send their messages through a public classic channel. Eve is able to read messages sent through the public classical channel, as well as manipulate quantum states sent through the quantum channel. In our protocol, Alice will send an encoded bit string to Bob which will eventually result in shared information which can be used to generate a shared key.

Protocol:

1. Alice chooses a random bit string (of 0's and 1's)
2. Alice randomly chooses an encoding basis (Rectilinear Basis or Diagonal Basis) for each bit
3. Using the encoding basis, Alice generates encoded quantum states for the bits and sends them to Bob
4. Bob randomly chooses a decoding basis (Rectilinear Basis or Diagonal Basis) for each bit
5. Using the decoding basis, Bob decodes the quantum states that Alice has sent. As a result, Bob now has a bit string as well (not necessarily the same as Alice's bit string)
6. Bob publicly states the decoding basis he used.

7. Alice checks the encoding basis Bob used for each bit with the decoding basis she used for each bit. If the basis used by both her and Bob are the same for a bit, Alice informs Bob that the bit is suitable for use. The bits for which different basis were used are discarded. The bits that are suitable for use are kept as shared information (key)
8. Bob reveals some of the bits to check for the presence of Eve
9. Alice confirms the bits to check whether eavesdropping occurred
10. The bits Bob revealed to check for the presence of Eve are discarded from the string of shared bits, and thus, the result is the shared secret key among Alice and Bob

Note: The protocol has been implemented [here](https://github.com/harismubashar/bb84_simulation) for simulation purposes.
(https://github.com/harismubashar/bb84_simulation)

Now that we have described the protocol, let's run through an instance of the protocol without the presence. Afterwards, we will run through another example where Eve does intercept the key distribution protocol.

5.1. Example (without Eve)

1. Alice randomly generates the bit string $b = 0111001110$
2. Alice randomly generate the basis $C_1 C_2 H_3 C_4 C_5 H_6 H_7 C_8 H_9 H_{10}$ where the subscript refers to the index used for the string b starting from 1. C is the rectilinear/computational basis and H is the diagonal/hadamard basis.
3. Thus, using her basis and string, Alice generates the states:

$$|\Psi_{00}\rangle |\Psi_{10}\rangle |\Psi_{11}\rangle |\Psi_{10}\rangle |\Psi_{00}\rangle |\Psi_{01}\rangle |\Psi_{11}\rangle |\Psi_{10}\rangle |\Psi_{11}\rangle |\Psi_{01}\rangle$$
and sends the states to bob.
4. Bob randomly generates the basis $H_1 C_2 H_3 H_4 C_5 C_6 H_7 C_8 H_9 C_{10}$
5. Bob uses his decryption basis with Alice's quantum states to get the bits:

$$b' = 0110011110$$
6. Bob states the decoding basis he used ($H_1 C_2 H_3 H_4 C_5 C_6 H_7 C_8 H_9 C_{10}$)
7. Alice confirms that the matching basis between her and Bob are at indexes 2, 3, 5, 7, 8, 9, which form the shared data their corresponding encoded bits
8. Bob releases bits encoded at index 3, 7, 8 to check for potential interception.
9. Alice confirms that Bob's bits at indexes 3, 7, 8 match her's.

10. Bits at index 3, 7, 8 are discarded, and thus only the bits at indexes 2, 5, 9 remain as shared data. Thus, the shared key among Alice and Bob is
- $$1_2 0_5 1_9 = 101$$

In this small example, we have gone through the protocol without any interception. As a result, Alice and Bob found on step 9 that all of the released bits match. We will now use the same example but add interceptor Eve right after step 3 when Alice sends her states to Bob.

5.2. Example (with Eve)

1. Alice randomly generates the bit string $b = 0111001110$
2. Alice randomly generate the basis $C_1 C_2 H_3 C_4 C_5 H_6 H_7 C_8 H_9 H_{10}$ where the subscript refers to the index used for the string b starting from 1. C is the rectilinear/computational basis and H is the diagonal/hadamard basis.
3. Thus, using her basis and string, Alice generates the states:

$$|\Psi_{00}\rangle |\Psi_{10}\rangle |\Psi_{11}\rangle |\Psi_{10}\rangle |\Psi_{00}\rangle |\Psi_{01}\rangle |\Psi_{11}\rangle |\Psi_{10}\rangle |\Psi_{11}\rangle |\Psi_{01}\rangle$$

and sends the states to bob.

- 1) Eve generates a random basis to decrypt Alice's states:

$$H_1 C_2 C_3 H_4 C_5 H_6 C_7 H_8 C_9 H_{10}$$

- 2) Eve measure the states using her basis and produces the following states to send to bob:

$$|\Psi_{01}\rangle |\Psi_{10}\rangle |\Psi_{10}\rangle |\Psi_{01}\rangle |\Psi_{00}\rangle |\Psi_{01}\rangle |\Psi_{10}\rangle |\Psi_{11}\rangle |\Psi_{10}\rangle |\Psi_{01}\rangle$$

4. Bob randomly generates the basis $H_1 C_2 H_3 H_4 C_5 C_6 H_7 C_8 H_9 C_{10}$
 5. Bob uses his decryption basis with Alice's quantum states to get the bits:
- $$b' = 0100001100$$
6. Bob states the decoding basis he used ($H_1 C_2 H_3 H_4 C_5 C_6 H_7 C_8 H_9 C_{10}$)
 7. Alice confirms that the matching basis between her and Bob are at indexes 2, 3, 5, 7, 8, 9, which form the shared data their corresponding encoded bits
 8. Bob releases bits encoded at index 3, 7, 8 to check for potential interception.
 9. Alice confirms that Bob's bits at indexes 3, 7, 8 match her's, but the bit at index 3 is 1 for Alice but 0 for Bob. This means there must have been an interception!

10. If Alice and Bob still wish to use the shared data for a key, the bits at index 3, 7, 8 are discarded, and thus only the bits at indexes 2, 5, 9 remain as shared data. Thus, the shared key among Alice and Bob is $1_2 0_5 1_9 = 101$

5.3. Case Analysis

This is the approach of the BB84 Protocol. Let us analyse what occurred when Eve had intercepted the key distribution. Specifically, let us focus on how Alice and Bob detect whether or not Eve had intercepted the photons that Alice had sent to Bob.

Let's assume that Eve intercepts the quantum channel and measures the encoded bits sent to Bob at Step 3. In order to measure the encoded bits, Eve must choose either a rectilinear or diagonal basis to filter the encoded bits. Since Eve has no knowledge of the bits, Eve must randomly choose the basis in order to detect what bits are being sent. Thus, for any single encoded bit, we are left with two possible cases:

- **Case 1:** Eve correctly determines the basis and is able to detect the correct encoded bit. Eve passes this bit on to Bob. In this case, Bob and Alice are unable to detect that Eve has intercepted the message.
- **Case 2:** Eve incorrectly determines the basis and determines the correctly encoded bit with 50% chance of failure. Eve passes this bit on to Bob.

Observe that on Step 7, Alice confirms the matching basis between herself and Bob. If Alice confirms that a basis for her and Bob matches, but the corresponding check bit (which Bob revealed on Step 8) does not match, then they can confirm that there must have been an interception by Eve because otherwise the quantum state of the qubit must have matched.

As we can see, this methodology to determine whether there is an interception or not is dependent on how many bits are revealed by Bob, which must be confirmed by Alice. Thus, Bob should reveal a sufficient amount of bits in order to determine whether Eve intercepted the message or not. Naturally, the more bits that are released by Bob the more sure that Alice and Bob can be that there has been no interception. Specifically, if all quantum states of the qubits match wherever the basis match, the probability that Eve *may* have intercepted can be stated as:

$$Pr[\text{Eve may have intercepted}] = \left(\frac{1}{2}\right)^n$$

since the probability of determining each basis correctly is $\frac{1}{2}$.

Let us analyse the percent error induced when Eve intercepts the message. In general, the percent error is determined by:

$$\%error = \left| \frac{E-T}{T} \right| \times 100$$

where E is the experimental value, and T is the theoretical value.

Let's consider the case where the total number of bits sent from Alice to Bob is $8n$ for some large $n > 1$. Then, half of the bits ($4n$) will be correctly determined by Bob on average, which can be used for the key (after discarding revealed bits). Note that if Eve did not intercept the message, then all of the bits with the same basis ($4n$ total bits) will match. Thus, our theoretical value is the number of bits that share basis, which is $4n$, and the experimental value is the bits which share basis and the quantum state, which is also $4n$. Thus, we have:

$$\%error = \left| \frac{4n-4n}{4n} \right| \times 100 = 0$$

so, in this scenario there is 0% percent error

Now, let's consider the case where Eve was to intercept the full message. In this case, Eve will also only be able to determine $4n$ bits of Alice's message on average. Eve will send $8n$ ($4n$ bits she correctly determined and $4n$ bits she incorrectly determined) to Bob, $4n$ of which Bob will share basis (with Alice) and $4n$ of which Bob will not share the basis (with Alice). Bob and Alice only consider the bits where they share their basis. Thus, the bits they share will be $\frac{4n}{2}$ which were not intercepted. And there will be $\frac{4n}{2}$ bits which were intercepted. The $\frac{4n}{2} = 2n$ bits which were not intercepted will have the correct bit values, however half of the bits (which is $\frac{2n}{2} = n$) which were intercepted will have incorrect bit values. Thus, Alice and Bob share basis with $\frac{8n}{2} = 4n$ bits and of these bits, n will contain incorrect bits values. Thus, the error rate detected from bits with similar basis will be:

$$\%error = \left| \frac{3n-4n}{4n} \right| \times 100 = 25\%$$

thus, in this scenario there is 25% percent error detected among bits with shared basis. Alice and Bob will abort the protocol in this scenario because they can determine that Eve must have intercepted the message.

We can see from this that if Eve intercepts the entire $8n$ length message sent by Alice, the error rate is expected to be 25% error rate from the bits that contain similar basis, which is n of the total number of bits sent by Alice at Step 1.

However, it is important to note that this is the ideal scenario. In this ideal scenario where no bits suffer from errors, a single checked bit which has a matching basis will indicate the presence of Eve, which will result in Alice and Bob aborting the key exchange. In practice, however, this is not necessary because bits may be corrupted due to environmental factors. As a result, in practice, Alice and Bob only keep the shared information if the probability of key bits being different is lower than an error threshold ϵ :

$$Pr[K_A \neq K_B] \leq \epsilon$$

Thus, the error rate will be modified depending on the correcting algorithms usable by Alice and Bob (depending on how many bits they are able to correct) and how sure they want to be that Eve is not eavesdropping.

6. Tensor Products

What we have discussed so far is relevant for a single qubit system. What about when we have a 2-qubit system or one larger? In order to understand the operations performed among two quantum state vectors $|\Psi\rangle$ and $|\Phi\rangle$, we must introduce tensor products. In mathematics, a tensor product a bilinear map represented by $\otimes$. Given two vector spaces, V and W , the tensor product $V \otimes W$ is a vector space which consists of all tensor products $v \otimes w$ where $v \in V$ and $w \in W$. For an example, if $V = \mathbb{R}^2$ and $W = \mathbb{R}^3$, then $v \in V$ can be represented by a two-dimensional vector basis $\{v_1, v_2\}$ and $w \in W$ can be represented by a three-dimensional vector basis $\{w_1, w_2, w_3\}$. An element in the

tensor product $V \otimes W$ can then be represented as $\{v_1 \otimes w_1, v_1 \otimes w_2, v_1 \otimes w_3, v_2 \otimes w_1, v_2 \otimes w_2, v_2 \otimes w_3\}$.

Specifically for Quantum Key Distribution, Tensor products are used for representing composite systems (such as representing a composite system of two Qubits) between vector spaces. For an example, the composite system of two bits A and B both being 0

is $|0_A 0_B\rangle = |0_A\rangle \otimes |0_B\rangle$. The tensor product essentially contains all the information of the individual bit vectors. All product operations done between any two vectors in this report are tensor products, even if the $\otimes$ is omitted, as there is no other reasonable way to create a product between the vectors while preserving their information.

To calculate the tensor product $v \otimes w$ where $v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ and $w = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$

we get:

$$v \otimes w = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \otimes \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} v_1 \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \\ v_2 \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} v_1 w_1 \\ v_1 w_2 \\ v_2 w_1 \\ v_2 w_2 \end{pmatrix}$$

For a qubit, we have the following vector representation:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

These vectors exist in a 2-dimensional Hilbert Space, thus only 2 basis vectors are required to represent the qubit states that live in this space.

Thus, for any two qubits in a composite system, we will have a 4-dimensional Hilbert Space because each of the vectors live in a different 2-dimensional Hilbert Space. We then have the 4 following basis vectors in this space:

$$|0\rangle \otimes |0\rangle = |00\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$|0\rangle \otimes |1\rangle = |01\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$|1\rangle \otimes |0\rangle = |10\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$|1\rangle \otimes |1\rangle = |11\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Observe how when we only considered one qubit, then our system, let's call it $|\Psi\rangle$ was represented as with two dimensions:

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

However, for a 2-qubit system, let's call it $|\Psi\rangle_2$ will be represented by using all the basis vectors:

$$|\Psi\rangle_2 = \alpha_1|00\rangle + \alpha_2|01\rangle + \alpha_3|10\rangle + \alpha_4|11\rangle$$

For a quantum system with more qubits, we would just repeat the process with another tensor product for each additional qubit.

7. No-cloning Theorem

Linearity in Quantum Channels

One extremely important postulate in Quantum physics, which we will use for the No-cloning theorem, is the linearity in quantum channels, which means that:

An operation L in a quantum channel is a linear map. Thus, for the same bit A and operation L , we have:

$$L(\alpha|p\rangle_A) = \alpha L(|p\rangle_A)$$

And

$$L(|p\rangle_A + |q\rangle_A) = L(|p\rangle_A) + L(|q\rangle_A)$$

$\Rightarrow$

$$L(\alpha|p\rangle_A + \beta|q\rangle_A) = \alpha L(|p\rangle_A) + \beta L(|q\rangle_A)$$

We will use this linearity of operations in quantum channels to prove the No-cloning Theorem. As a side note, one notable consequence of this linearity is that Quantum channels can be represented by completely positive and trace-preserving maps.

No-cloning Theorem Proof

The No-cloning theorem is an extremely important concept which shows that it is impossible to clone arbitrary quantum states (and also show that it is only possible to create a copy in classical states).

We know the following properties:

$$|\Psi\rangle_A = \alpha|0\rangle_A + \beta|1\rangle_A \quad (1)$$

We know this as the superposition distribution of the a bit at index A.

We also know the “clone” operation:

$$L(|\Psi\rangle_A |0\rangle_B) = |\Psi\rangle_A |\Psi\rangle_B \quad (2)$$

This is the clone operation that operation L performs. It Takes the first bit at index A, namely $|\Psi\rangle_A$, and copies it to the bit at index B, converting it from its previous state of $|0\rangle_B$ to $|\Psi\rangle_B$, essentially making a clone of $|\Psi\rangle$ because there are two instances of $|\Psi\rangle$ after the L operation (one at index A, and one at index B)

Thus, by (2), we have:

$$\begin{aligned} & L(|\Psi\rangle_A |0\rangle_B) \\ &= |\Psi\rangle_A |\Psi\rangle_B \\ &= (\alpha|0\rangle_A + \beta|1\rangle_A)(\alpha|0\rangle_B + \beta|1\rangle_B) \end{aligned}$$

by substituting (1)

$$\begin{aligned} &= (\alpha|0\rangle_A \alpha|0\rangle_B + \alpha|0\rangle_A \beta|1\rangle_B + \beta|1\rangle_A \alpha|0\rangle_B + \beta|1\rangle_A \beta|1\rangle_B) \\ &= \alpha^2|0\rangle_A |0\rangle_B + \alpha\beta|0\rangle_A |1\rangle_B + \beta\alpha|1\rangle_A |0\rangle_B + \beta^2|1\rangle_A |1\rangle_B \\ &= \alpha^2|00\rangle + \alpha\beta|01\rangle + \beta\alpha|10\rangle + \beta^2|11\rangle \quad (A) \end{aligned}$$

Note that we can also write $|\Psi\rangle_A |0\rangle_B$ as follows:

$$|\Psi\rangle_A |0\rangle_B$$

$$= (\alpha|0\rangle_A + \beta|1\rangle_A)|0\rangle_B$$

by substituting **(1)**

$$\begin{aligned} &= \alpha|0\rangle_A|0\rangle_B + \beta|1\rangle_A|0\rangle_B \\ &= \alpha|00\rangle + \beta|10\rangle \text{ **(3)** } \end{aligned}$$

Thus, we have:

$$\begin{aligned} &L(|\Psi\rangle_A|0\rangle_B) \\ &= L(\alpha|00\rangle + \beta|10\rangle) \end{aligned}$$

by substituting **(3)**

$$\begin{aligned} &= L(\alpha|00\rangle) + L(\beta|10\rangle) \\ &= \alpha L(|00\rangle) + \beta L(|10\rangle) \end{aligned}$$

by using the linearity of quantum operations mentioned previously

$$\begin{aligned} &= \alpha|00\rangle + \beta|11\rangle \\ &= \alpha|00\rangle + 0|01\rangle + 0|10\rangle + \beta|11\rangle \text{ **(B)** } \end{aligned}$$

Thus we see that **(A)** and **(B)** give us gives us:

$$\begin{aligned} \text{**(A)**} &= \text{**(B)**} \\ \alpha^2|00\rangle + \alpha\beta|01\rangle + \beta\alpha|10\rangle + \beta^2|11\rangle &= \alpha|00\rangle + 0|01\rangle + 0|10\rangle + \beta|11\rangle \\ &\Rightarrow \\ \alpha^2 &= \alpha \\ \beta^2 &= \beta \\ \alpha\beta &= 0 \text{ or } \beta\alpha = 0 \end{aligned}$$

What is interesting is that $\alpha\beta = 0$ or $\beta\alpha = 0$, which means that either $\alpha = 0$ or $\beta = 0$, which are precisely the cases where $|\Psi\rangle_A$ is in a classical state and no superposition distribution is present. Thus, we can see that this equality cannot be satisfied when $|\Psi\rangle_A$ is in a superposition and there are non-zero values of α and β , implying that cloning arbitrary quantum states is impossible.

Note: All the multiplications done in this theorem consists of tensor products. They have been omitted so it is easier to follow the proof.

8. Security

8.1 Quantum Resilience

Let us take a look at the rationale behind quantum algorithms which are theoretically able to solve current, classical cryptographic schemes by utilizing the power of quantum computing. The two algorithms with these implications are Shor's algorithm and Grover's algorithm. Let's take a look at each of these algorithms and see why the BB84 protocol is immune to them.

Shor's Algorithm and Grover's Algorithm

Shor's algorithm is a quantum algorithm which is designed to factor large composite numbers to their prime factors and computing discrete logarithms. As a result of employing this algorithm, various cryptographic schemes can be solved quickly. For example, Shor's algorithm can factor the product of large prime numbers in polynomial time, which breaks the security of RSA. Elliptic Curve Cryptography algorithms are also vulnerable because Shor's algorithm can compute discrete logarithms in elliptic curve groups.

Shor's algorithm utilizes quantum parallelism in quantum computers to solve these problems exponentially faster than classical computers. As a brief explanation, quantum parallelism is the concept by which quantum computers can compute multiple computations simultaneously using quantum states, as opposed to sequential computation in classical computers.

Grover's algorithm is another quantum algorithm which is designed to provide a significant speedup for search algorithms. It is a quantum search algorithm which can search an unsorted database in $O(\sqrt{N})$ time where N is the size of the domain. In the context of cryptography, what this implies is that brute force algorithms which are designed to determine an input will need to only find $\sqrt{N}$ length, thus reducing the computational time required to break encryptions such as AES or hash functions.

It is easy to see that the BB84 protocol is not vulnerable to Shor's algorithm because the protocol is not dependent on the computational complexity of number theory problems such as factoring primes or computing discrete logarithms. Instead, the security of the BB84 protocol lies in the fact that arbitrary quantum states cannot be

cloned. Thus, an eavesdropper cannot fundamentally clone a quantum state in order to forward it.

We can also see that Grover's algorithm will not help in breaking BB84 because there is no encryption algorithm pattern which can be found or broken in order to find the message. This is because the security of BB84 lies in the fact that choosing the correct basis is completely random and not due to any type of methodological or computational pattern. Thus, Grover's algorithm could not be used to break BB84.

8.2 Proof approach for BB84

Although we can prove the security of BB84 theoretically, it is important to note that in practice, there are other complications which must be taken into account, such as the reliability of measurement tools used, which are often not perfect.

Security Requirements:

1. Alice and Bob's should have the same keys

The probability that Alice and Bob's key differ but the protocol does not abort is small:

$$Pr[K_A \neq K_B] \leq \epsilon$$

2. The key is uniformly random and Eve should have no knowledge of the Key

In order to prove the security of the BB84 protocol, we must introduce the entanglement based BB84 protocol. The reason for this is that the entanglement version of BB84 protocol can be reduced to the prepare-and-measure version of BB84 protocol (the one we have described in detail in the report). Thus, proving the security of the entanglement based BB84 protocol will imply the security of the prepare-and-measure version of BB84 protocol as demonstrated by Peter W. Shor and John Preskill.

Maximally Entangled States

To create an entanglement based protocol, it is essential to use maximally entangled states. Maximally entangled states are the quantum states which have the highest amount of entanglement possible for a quantum system. The important detail here is that two entangled particles cannot be described independently. This is important because if two parties share a maximally entangled state, then a third party cannot have any entanglement with it, and thus, the bit in the maximally entangled state is perfectly

secure. For qubits, which have two basis states (one for each bit), there are four possible maximally entangled states, known as Bell states, as follows:

For a pair of qubits, the two most common maximally entangled states are the Bell states. There are four Bell states, and they can be expressed as follows:

- State $|\Phi^+\rangle$ which represents the superposition of the two basis states $\{|00\rangle, |11\rangle\}$
- State $|\Phi^-\rangle$ which represents the superposition of the two basis states $\{|00\rangle, |11\rangle\}$ but with a different phase relation
- State $|\Psi^+\rangle$ which represents the superposition of the two basis states $\{|01\rangle, |10\rangle\}$
- State $|\Psi^-\rangle$ which represents the superposition of the two basis states $\{|01\rangle, |10\rangle\}$ but with a different phase relation

Understanding what phase relations are or how the bell states are represented are not important for understanding the protocol. All we need to know is that if two parties share a maximally entangled state, then a third party cannot have any entanglement with it.

In the setup for entanglement based protocols, Alice and Bob have access to a source which can distribute quantum states. Eve can also use this source. For the sake of the protocol, we will assume that Eve has access to the source. Just like the prepare-and-measure based version, Eve can listen to the classical channel, but cannot modify the messages there.

For the protocol, Alice and Bob want to distribute maximally entangled states between them which can be used for generating a key.

Protocol:

1. Alice prepares $2n$ qubit pairs which are in the maximally entangled state $|\Phi^+\rangle$ (each pair is maximally entangled)
2. Alice selects n qubits which will be used to check for errors in the qubit pairs
3. Alice creates a message m of bit length $2n$ and applies a Hadamard transformation to her half of the pair. She applies the transformation whenever a bit m is 1.
 - Hadamard transformation is a transformation of a quantum state to a superposition. This transformation is its own inverse, so using a Hadamard transformation on the state again will return the original qubit state.
4. Alice sends the other qubit pairs she has not applied the Hadamard transformation to Bob.

5. Bob receives the qubit pairs designated for him and broadcasts that he has the qubits
6. Alice announces the message m
7. Bob applies the Hadamard transformation on his bits
8. Both Alice and Bob check the qubits in the basis $|1\rangle$ and $|0\rangle$ to determine the error rate. If the error is above threshold, then they abort the protocol.

The remaining proof is quite complex and requires a thorough understanding of quantum physics and quantum entanglement to grasp. Thus, it will be skipped as it is lengthy and out of scope of this paper. For reference, the proof by Peter W. Shor and John Preskill is [here](https://arxiv.org/pdf/quant-ph/0003004.pdf) (<https://arxiv.org/pdf/quant-ph/0003004.pdf>). Instead, we can discuss why an entanglement version of BB84 would be secure.

Since we have established that the bits generated by Alice on Step 1 are maximally entangled with the bits which are sent to Bob, it follows that the bits cannot be interfered by Eve without being instantly noticeable by either of them. Thus, if Alice and Bob share n bits which are maximally entangled, they can use the entirety of the n bits as shared data without the risk of Eve being able to detect the quantum states of the qubits. From this type of situation, it is easy to see why the entanglement version is secure.

9. Conclusion

The BB84 is an incredible advancement in the realm of key distribution protocols which utilizes the principles of quantum mechanics in order to ensure secure communication which are resilient to the potential challenges that classical protocols will face in the future. Although the intricate details are complex to understand, the algorithm itself is quite easy to follow through and analyse. Unlike classical algorithms, BB84 is able to detect the presence of interceptors with an extremely high success rate, which is unprecedented in classical protocols. However, there are difficulties faced for the practical implementation of the protocol. Creating world-wide systems which use accurate photon detectors throughout is a considerable technical challenge, and one we must face if algorithms which utilize photon encoding are to be wide-spread. Even though BB84 is proven to be immune to evesdroppers, it is not immune to all practical threats, such as manipulation of photon detectors. Although BB84 is an introduction to quantum key distribution and many advancements have come regarding QKD protocols, such as entanglement-based protocols, BB84 serves as a crucial foundation for key distribution algorithms to come.

10. References

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